A Comparative Study of Regression, Classification, and Clustering for Practical Machine Learning

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***Abstract*—This paper reports three practical machine-learning investigations: walking-speed regression using BLE-GSpeed, binary network intrusion detection using NSL-KDD, and unsupervised product-title clustering. The projects required different preprocessing and evaluation strategies. BLE-GSpeed contains approximately 910,780 observations but only about 382 labelled walking sessions, with one speed label per session. Initial window-level models performed poorly; session-level aggregation produced a strong tuned Random Forest validation result (MAE=0.01302, RMSE=0.01880, $R^2 = 0.99617$). For NSL-KDD, Logistic Regression, KNN, and Decision Tree were compared using stratified train-validation-test splits, training-only scaling, and SMOTE on training data only. Decision Tree achieved 99.54% test accuracy and 99.51% attack F1, while KNN achieved the highest ROC-AUC and PR-AUC but slower prediction. For product clustering, TF-IDF represented 35,311 titles using 37,178 terms. K-Means++ with $K = 14$ achieved the highest sampled cosine silhouette score among tested values (0.0489); DBSCAN found 273 clusters and marked 46.71% of a 10,000-title sample as noise. These findings highlight the importance of matching representations to label structure, preventing leakage, and interpreting metrics critically.**

I. Introduction

This project applies regression, supervised classification, and unsupervised clustering to three practical datasets. The aim is to develop reproducible preprocessing pipelines, compare candidate methods using appropriate metrics, and discuss limitations. The results are not directly rankable across tasks because each has a different objective and evaluation criterion.

II. Datasets

A. BLE-GSpeed

The dataset contains approximately 910,780 rows and eight columns: `mac`, `rssi`, `device`, `timestamp`, `user`, `direction`, `walk_id`, and `speed`. Measurements are grouped into about 382 walking sessions, each with one speed label. The task predicts a continuous speed value.

B. NSL-KDD

The task is binary classification: normal connections are mapped to 0 and attack connections to 1. Logistic Regression, K-Nearest Neighbours (KNN), and Decision Tree are compared. Attack recall is important because a false negative is an attack incorrectly labelled normal.

C. Product titles

The product dataset contains 35,311 rows and seven columns: Product ID, Product Title, Merchant ID, Cluster ID, Cluster Label, Category ID, and Category Label. No missing cells or exact duplicate rows were reported. There are 30,993 unique title strings and 3,184 distinct titles occur more than once. Only Product Title is used as clustering input; existing cluster/category fields are reserved for post-hoc interpretation.

III. Preprocessing and Feature Engineering

A. BLE-GSpeed

RSSI varies with distance, device orientation, body obstruction, multipath propagation, interference, hardware, and environment. After chronological sorting, missing RSSI values were forward-filled only within the same user, session, and device. This avoids propagating measurements across unrelated sessions. The initial workflow aggregated irregular observations into 500-ms windows and extracted RSSI statistics (mean, median, standard deviation, minimum, maximum, range, and counts where available). Temporal features such as lagged values, changes, and trailing summaries were explored within sessions without using future observations. Raw identifiers and timestamps were not treated as direct predictors.

The principal difficulty was a mismatch between observation and label units: many correlated BLE measurements share a single session-level speed label. Treating every observation as independent can inflate the apparent sample size and create leakage if windows from one walk are split across partitions. We therefore aggregated window features into one feature row per labelled session. This better matches the data-generating process and reduces short-term signal noise. However, session-level estimation is not equivalent to real-time prediction from a single short window.

B. NSL-KDD

Column names were stripped of whitespace, lowercased, and cleaned of trailing punctuation. The workflow drops columns containing any missing values and then removes constant columns. It removes the same eight correlated features as the earlier experiment: `num_root`, `srv_serror_rate`, `srv_rerror_rate`, `dst_host_srv_serror_rate`, `dst_host_serror_rate`, `dst_host_rerror_rate`, `dst_host_srv_rerror_rate`, and

TABLE I
BLE-GSPEED PREPROCESSING: BEFORE AND AFTER

Stage	Before	Treatment / After	Purpose
Raw data	∼910,780 rows, 8 columns	Schema and quality checks; raw rows retained as source	Establish structure
Missing RSSI	Missing readings possible	Forward-fill within chronological user/session/device group; unresolved leading gaps handled downstream	Avoid cross-session contamination
Temporal representation	Irregular observations	Initial 500-ms windows with RSSI summaries	Reduce irregularity/noise
Temporal features	Window summaries	Lagged values, changes and trailing summaries tested within sessions	Capture dynamics without future data
Model input	Repeated windows share one speed label	Session aggregation, about 382 labelled examples	Match feature and label units
Validation	Related windows may cross partitions under random row split	Keep sessions intact across partitions; fit transforms on training only	Reduce optimistic estimates

`dst_host_same_srv_rate`. `protocol_type` was mapped as ICMP=0, TCP=1, UDP=2; `flag` retained the original mapping; and `service` was dropped.

A stratified 70/15/15 train-validation-test split was used. Median imputation and MinMax scaling were fitted on training data only. SMOTE was applied only to scaled training data; validation and test distributions remained unchanged. This avoids leakage from fitting a scaler or creating synthetic examples before splitting. Integer mappings for protocol and flag preserve the prior workflow but impose artificial ordering; one-hot encoding is a possible future comparison.

C. Product-title text

Column whitespace was stripped. Repeated titles were retained because different merchants can list the same item; there were no exact duplicate rows. Product IDs, merchant IDs, existing cluster labels, and category labels were excluded from model input to prevent metadata leakage. TF-IDF used lowercasing, accent stripping, word unigrams and bigrams, minimum document frequency 2, maximum document frequency 0.98, sublinear term frequency, and L2 normalisation. The matrix shape was $35{,}311 \times 37{,}178$ with density 0.000409 (about 0.041%). Unigrams capture terms such as "iphone" and "64gb", while bigrams capture phrases such as "iphone 8" and "washing machine". For DBSCAN, a 10,000-title sample was reduced to 50 dimensions using TruncatedSVD and L2-normalised; the summed explained variance ratio was 0.1676.

The main text limitation is lexical variation: the same product may differ in spacing, abbreviations, model codes, colour names, or language (e.g., "64gb" versus "64 gb"). TF-IDF is interpretable but does not understand semantic equivalence. Character n-grams and domain-specific normalisation are possible extensions.

IV. EXPERIMENTS AND RESULTS

A. BLE-GSpeed Regression

Linear Regression was used as a linear baseline, Random Forest as a nonlinear ensemble, and a mean predictor as a reference. Initial window-level models had negative R^2, meaning they performed worse than a mean predictor on that split. Adding temporal features and tuning Random Forest improved the later window-level test R^2 to 0.09437, but performance remained limited. Session-level aggregation produced much stronger validation results.

The session-level Random Forest is the best current validation candidate. The large improvement must be checked for target leakage, target-derived features, and session overlap. Final held-out test metrics for this experiment were not supplied; therefore, the validation score is not proof of generalisation. Testing on unseen users would provide a stricter evaluation.

B. NSL-KDD Classification

Logistic Regression tested $C \in \{0.1, 1, 10\}$. KNN tested $k \in \{3, 5, 7, 11\}$ with uniform and distance weighting. Decision Tree tested depths None, 10, 20, and 30 with minimum leaf sizes 1, 5, and 10. Selection prioritised attack F1 and then attack recall. Selected models were refitted on train plus validation data; test data remained untouched.

Decision Tree achieved the best threshold-based test results: 99.54% accuracy, 99.52% attack recall, and 99.51% attack F1. KNN was close on these metrics and had the highest ROC-AUC (0.9982) and PR-AUC (0.9973), but its measured prediction time was 1.1853 s compared with 0.0014 s for Decision Tree. Logistic Regression prediction time was 0.0006 s, but attack recall was lower at 94.02%. Its confusion matrix contained 9,729 true normal predictions, 8,269 correctly detected attacks, 372 false positives, and 526 false negatives. These results are dataset-specific and do not guarantee performance on live network traffic.

C. Product Clustering

K-Means++ was evaluated for $K = 2$ to 15. Inertia decreased from 34,715.357 at $K = 2$ to 32,807.997 at $K = 15$, but the curve had no decisive elbow. The best sampled cosine silhouette was 0.0489 at $K = 14$.

The low silhouette values indicate overlapping clusters. Some groups were coherent (e.g., multiple Bosch microwave

TABLE II
NSL-KDD PREPROCESSING: BEFORE AND AFTER

Stage	Before	Treatment / After	Purpose / Caveat
Target	Normal and multiple attack labels	Normal=0; non-normal=1	Binary IDS task
Missing/constant fields	Potential missing and constant columns	Drop columns containing missing values; remove constants	Column-wise deletion
Feature reduction	Standard connection features	Remove eight correlated features and drop service	Preserve earlier choices
Categorical fields	Protocol and flag strings	Original integer mappings retained	Artificial order may be imposed
Data split	Full cleaned matrix	Stratified 70/15/15 train/validation/test	Keep final test held out
Imputation/scaling	Different scales and possible unmapped values	Median imputer and MinMaxScaler fitted on training only	Prevent leakage; supports KNN
Class imbalance	Original training distribution	SMOTE on training data only	Held-out sets remain unchanged

TABLE III
PRODUCT-TITLE PREPROCESSING

Stage	Before	Treatment / After
Dataset	35,311 rows, 7 columns	Column names stripped of whitespace
Quality	Repeated titles	0 missing cells, 0 exact duplicate rows; 30,993 unique title strings
Model input	IDs, title and reference labels	Product Title only
Text features	Raw strings	TF-IDF with 37,178 terms; 0.041% density
DBSCAN input	37,178-dimensional TF-IDF	10,000-title sample reduced to 50 SVD dimensions

TABLE IV
BLE-GSPEED REGRESSION RESULTS

Model / Experiment	MAE	RMSE	R^2	Within ± 0.1
Initial window LR (test)	0.20506	0.26440	-0.01178	35.04%
Initial window RF (test)	0.21132	0.26895	-0.04689	32.54%
Tuned window RF (test)	0.18939	–	0.09437	–
Session LR (validation)	0.08948	0.11623	0.85355	63.79%
Session tuned RF (validation)	0.01302	0.01880	0.99617	100.00%
Session mean baseline (validation)	0.24374	0.30378	-0.00037	31.03%

TABLE V
NSL-KDD VALIDATION RESULTS

Model	Acc.	Prec.	Rec.	F1	ROC	PR
Decision Tree	0.9952	0.9950	0.9948	0.9949	0.9952	0.9922
KNN	0.9944	0.9939	0.9941	0.9940	0.9973	0.9959
Logistic Regression	0.9533	0.9554	0.9437	0.9495	0.9883	0.9887

TABLE VI
NSL-KDD HELD-OUT TEST RESULTS

Model	Acc.	Prec.	Rec.	F1	Bal. Acc.	ROC	PR
Decision Tree	0.9954	0.9949	0.9952	0.9951	0.9954	0.9957	0.9927
KNN	0.9951	0.9944	0.9950	0.9947	0.9951	0.9982	0.9973
Logistic Regression	0.9525	0.9569	0.9402	0.9485	0.9517	0.9880	0.9883

listings), while others mixed appliance types. DBSCAN on the 10,000-title reduced sample found 273 non-noise clusters and marked 4,671 records (46.71%) as noise. The large noise share and many small clusters suggest parameter sensitivity and a representation that does not yet yield compact density groups. Post-hoc category-label NMI was reported as 0.6326 for K-Means and 0.3964 for DBSCAN; these indicate association with broad categories, not proof of exact-product deduplication. Confirm these values against the executed notebook before submission.

TABLE VII
K-MEANS++ ELBOW AND SILHOUETTE RESULTS

K	Inertia	Silhouette
2	34715.357	0.0117
3	34411.202	0.0173
4	34174.580	0.0228
5	34014.350	0.0260
6	33786.178	0.0322
7	33674.998	0.0342
8	33512.972	0.0343
9	33413.989	0.0375
10	33267.776	0.0393
11	33177.116	0.0414
12	33060.866	0.0426
13	32929.256	0.0418
14	32831.701	0.0489
15	32807.997	0.0472

TABLE VIII
CLUSTERING COMPARISON

Method	Result	Interpretation
K-Means++ ($K = 14$)	Silhouette = 0.0489; no clear elbow	Complete partition, but weak separation
DBSCAN	273 clusters; 46.71% noise	Finds local groups but fragmented

V. Cross-Task Discussion

The largest BLE improvement followed alignment of features with the session-level target, but test evaluation remains outstanding. For NSL-KDD, Decision Tree had the strongest default-threshold detection and low prediction time, while KNN ranked probabilities best by ROC-AUC and PR-AUC at a higher inference cost. For product clustering, TF-IDF provided a scalable, interpretable baseline, but low silhouette and high DBSCAN noise indicate that more work is needed.

VI. Limitations and Future Work

For BLE-GSpeed, run the final session-level test evaluation, audit features for target or post-outcome leakage, compare 250-ms, 500-ms, and 1,000-ms windows, and test on unseen users. For NSL-KDD, retain the untouched test set, consider one-hot encoding, inspect per-class errors, and evaluate on independent network data. For product clustering, normalise model codes, units, colours, punctuation, and multilingual variants; test character n-grams; tune K, DBSCAN `eps` and `min_samples`; and inspect representative titles. Across tasks, record data versions, random seeds, library versions, and exact preprocessing settings.

TABLE IX
Summary of Best Available Evidence

Task	Best Evidence and Conclusion
BLE-GSpeed	Tuned Random Forest: validation MAE = 0.01302, RMSE = 0.01880, and $R^2 = 0.99617$. Promising validation result, but held-out testing is still required.
NSL-KDD	Decision Tree: test accuracy = 0.9954 and F1 = 0.9951. KNN achieved ROC-AUC = 0.9982 and PR-AUC = 0.9973. Decision Tree leads on threshold-based metrics and speed; KNN leads on ranking metrics.
Product titles	K-Means++ ($K = 14$): silhouette = 0.0489. DBSCAN: 46.71% noise. The results are exploratory; representation and clustering parameters need improvement.

VII. Conclusion

This paper presented three machine-learning studies. BLE-GSpeed showed that matching aggregation to the session-level target substantially improved validation, though held-out testing remains necessary. In NSL-KDD, leakage-aware scaling and training-only SMOTE supported a fair comparison; Decision Tree had the strongest threshold-based test results, while KNN had the highest ROC-AUC and PR-AUC. Product-title clustering showed that TF-IDF supports exploratory clustering, but low silhouette and high DBSCAN noise reveal substantial overlap. Appropriate preprocessing, validation, and cautious interpretation are essential across all three tasks.

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